

SCORES

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The researcher is comparing students' scores using two different methods and wants to know if there are any differences. Two independent groups of students took the same standardized test.

```
##   group_A group_B
## 1      78      70
## 2      85      72
## 3      88      75
## 4      90      80
## 5      92      83
```

Table 1: number of students of each group

group_A	group_B
78	70
85	72
88	75
90	80
92	83

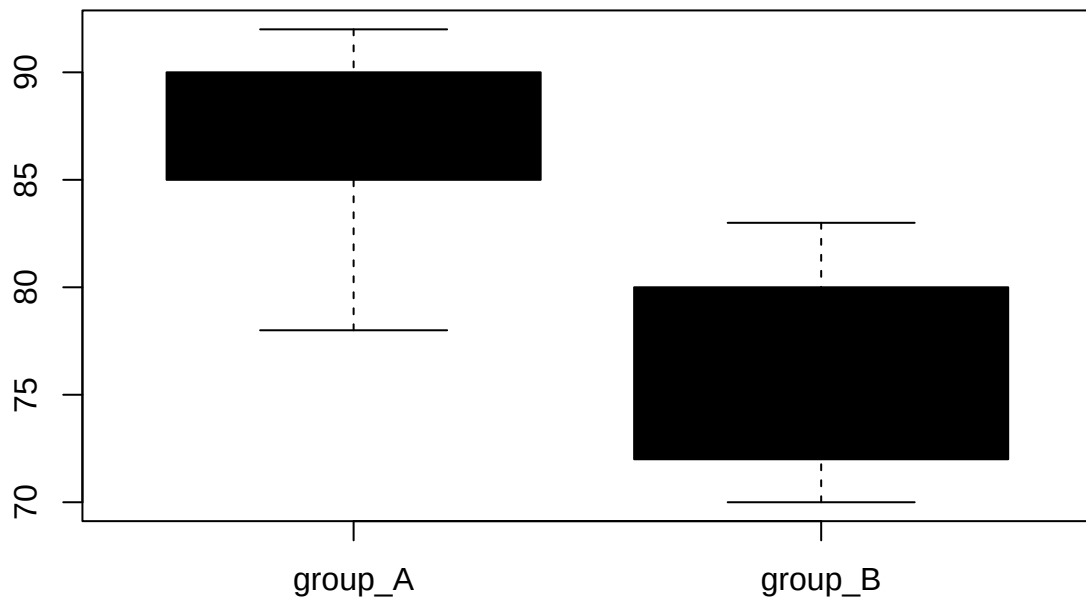
Table one: show number of students for each group, and number of students are the same .

Descriptive statistic :

Table 2: Descriptive statistic for group_A and group_B

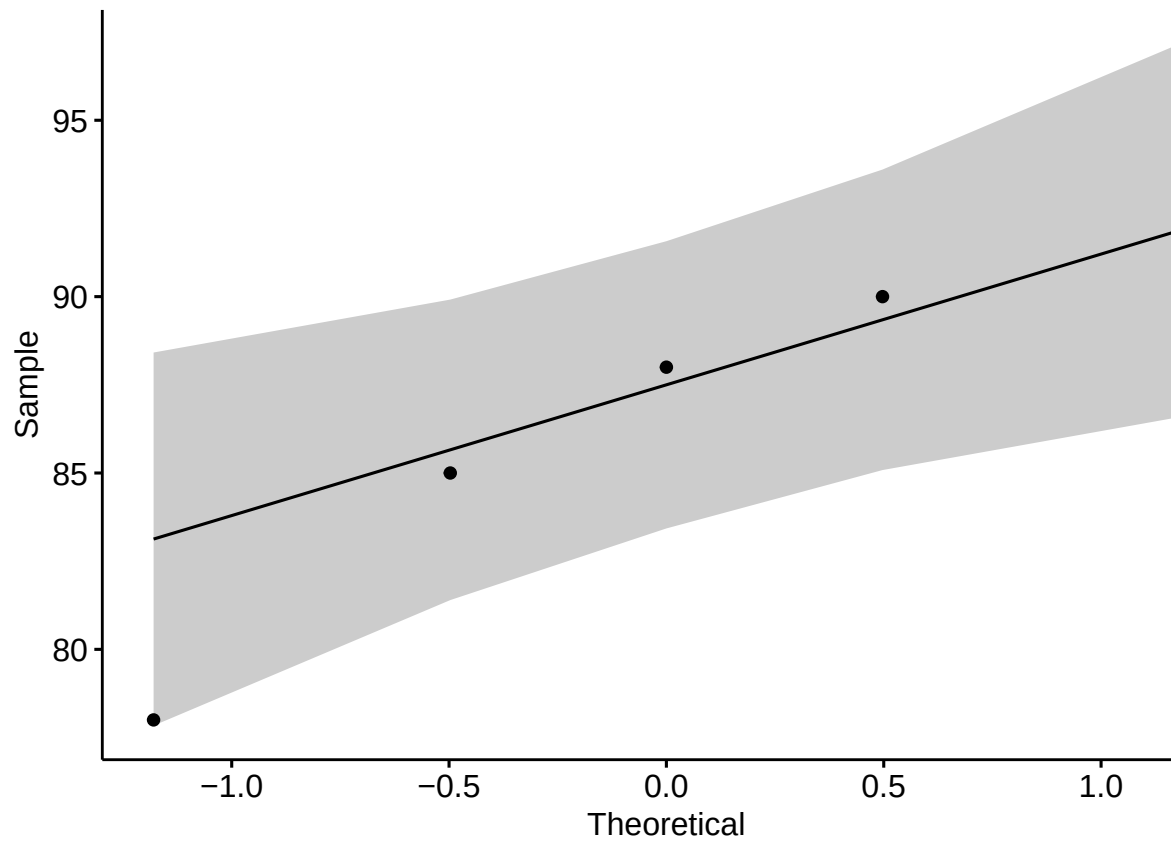
	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
group_A	1	5	86.6	5.46	88	86.6	4.45	78	92	14	-0.54	-1.54	2.44
group_B	2	5	76.0	5.43	75	76.0	7.41	70	83	13	0.16	-2.03	2.43

The mean of group_A is 86.6 and standard deviation is 5.46 ,the mean of group_B is 76 and standard deviation is 5.43 ,mean of group_A is greater than mean of group_B ,the approximate same dispersion ,and Highest grade in Group A is 92,the less grade is 78 ,and the Highest grade in group_B is 83 , the less grade is 70 .

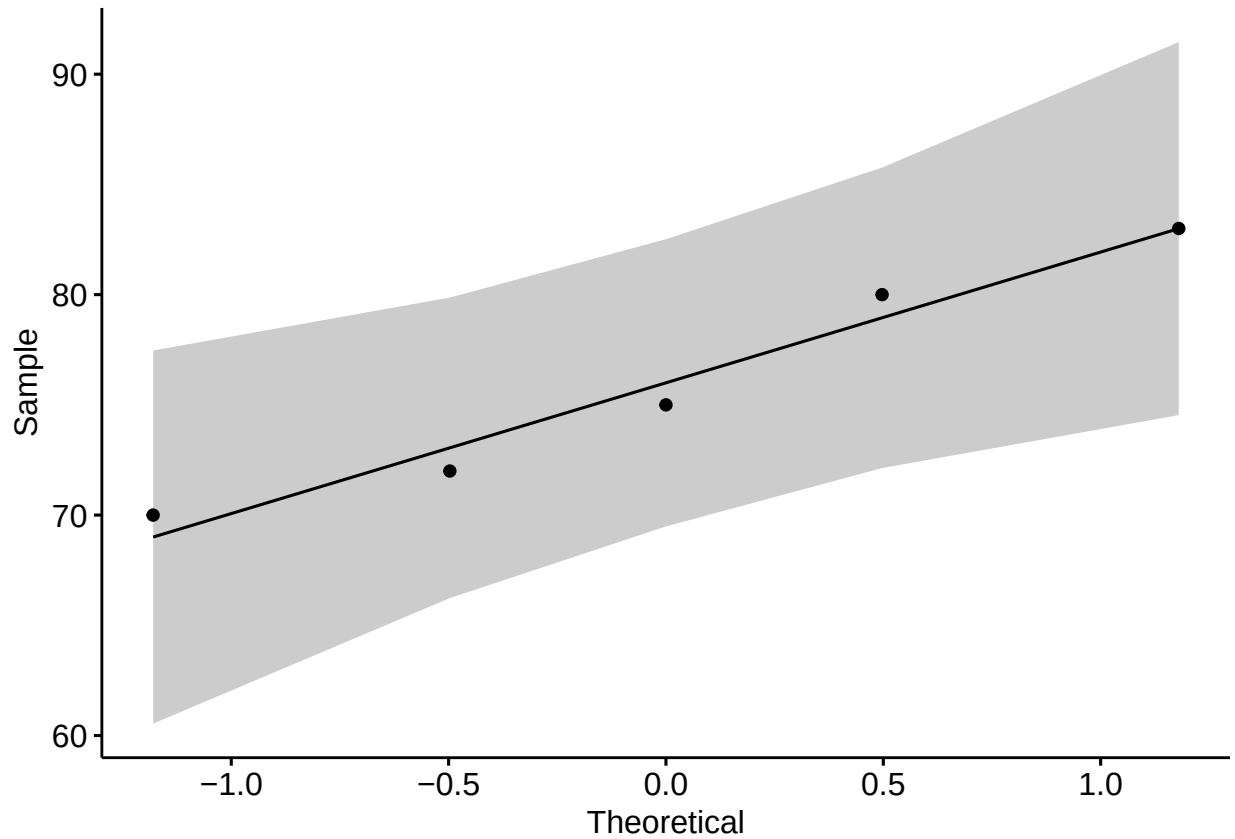


The graph showed that group_ A achieved higher grades than group _B.

Inference statistic :



check the assumption :



The results showed that the group's scores were not normally distributed.

therefore, the Wilcoxon signed-rank test was used to study the difference between the two groups, and the results showed the following:

```
##
## Wilcoxon rank sum exact test
##
## data: group_A and group_B
## W = 23, p-value = 0.03175
## alternative hypothesis: true location shift is not equal to 0
```

There are statistically significant differences where the value reached $p\text{-value}=0.03175$ less than 0.05 ,This means there are differences in grades between group_A and group_B.